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K_AUD = SQRT(2) x LN(2): THE CROSSING ZONE HINGE
Three Thresholds and the Identity $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$

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ABSTRACT

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K_AUD = $\sqrt{2} \times \ln(2) \approx 0.980258$ is the framework's central constant – geometry's simplest diagonal times information's simplest cost. Its complement relative to unity is the structural gap $G = 1 - K_AUD \approx 0.019742$ (about 2%). This paper zooms in on the 35-37 step interval of the Binary Tower – the framework's measurement tool – and locates three exactly-positioned thresholds within it, together with the algebraic identities that bind their spacings into a single structural object.

The Binary Tower is a cumulative function $n \cdot G$ evaluated at integer steps $n \in [0, 64]$, where G is the gap defined above. Its step definition is established in Paper 4 (Gap Scaling Across Domains); its self-similarity formula in Paper 3 (§7.2). At each integer step n , the Tower returns $n \cdot G$ as a dimensionless quantity. For analysis of the threshold crossings addressed here – which occur at noninteger positions – we work with the cumulative function extended continuously as $T(x) = x \cdot G$ for real $x \in [0, 64]$; the Binary Tower's integer steps are the marked values $T(n)$, $n \in \mathbb{Z}$, and each threshold n_i is the exact real solution of $T(x) = c_i$ for a target constant c_i .

The answer, in the $[35, 37]$ interval, is three thresholds – Landauer, geometric damping, Shannon bound – and a binding identity that ties their spacings to K_AUD itself.

1. FRAMEWORK IN BRIEF

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The framework's central constant is $K_AUD = \sqrt{2} \times \ln(2) \approx 0.980258$, the product of one geometric primitive ($\sqrt{2}$) and one informational primitive ($\ln 2$). Its complement relative to unity is the framework's structural gap $G = 1 - K_AUD \approx 0.019742$, derived and verified in Paper 3 (Complete Framework v3.3.1, OSF DOI 10.17605/OSF.IO/QH5S2). The gap is irreducible: G is transcendental – the difference $1 - \sqrt{2} \cdot \ln(2)$ of an algebraic and a transcendental quantity – and no nonzero integer multiple of G equals any algebraic number (Hermite-Lindemann).

The three thresholds are: the Landauer threshold ($n \cdot G = \ln 2$), the geometric damping threshold ($n \cdot G = 1/\sqrt{2}$), and the Shannon bound ($n \cdot G = 1/(2 \ln 2)$), clustered in the integer interval $[35, 37]$. The remainder of the paper develops these in turn (§§2-4), proves the algebraic identity binding them (§5), places them within the framework's broader architecture (§6), addresses why three thresholds and not others (§7), and closes with tier classification, verification, and failure conditions (§8).

2. APPEARANCE 1: THE LANDAUER THRESHOLD

The first of the three thresholds in the crossing zone is the position at which the Binary Tower's cumulative gap first reaches the thermodynamic cost of erasing a single bit of information. From the relation

$$n \cdot G = \ln 2,$$

the threshold position is

$$n_L = \ln 2 / G \approx 35.111$$

(precise at mpmath dps=100: $n_L = 35.11054\dots$). The threshold is non-integer and lies in the open interval (35, 36); the Tower's cumulative value $T(n) = n \cdot G$ crosses $\ln 2$ as n passes through this interval, at exactly n_L . (§5c strengthens "non-integer" to *irrational* – a structural consequence of G 's transcendence.)

Mechanism.

The value $\ln 2$ is the information content of one bit expressed in nats. Landauer's principle assigns the irreversible erasure of one binary distinction a minimum thermodynamic cost of $k_B \cdot T \cdot \ln 2$ at temperature T (Landauer 1961); in the Tower's dimensionless framework units ($k_B \cdot T = 1$), this cost equals $\ln 2$. The Landauer threshold is the position where the Tower's cumulative value first equals this bit-erasure cost.

Below n_L , the cumulative value $T(n) = n \cdot G$ is sub-Landauer: the framework's stacked gap has not yet reached the cost of one bit erasure. For $n > n_L$, the cumulative value $T(n) = n \cdot G$ exceeds $\ln 2$. The threshold thus marks an information-theoretic boundary inscribed into the Tower's geometry by the foundational principle of binary computation.

Atlas lock.

`landauer\_crossing` – Theorem-tier; value $n \cdot G = \ln 2$; position $n_L \approx 35.111$.

Verification.

mpmath dps = 60 corpus pass, 2026-06-29; independent re-verification at dps = 100, 2026-06-29 (residual 10^{-100} , mpmath precision limit). The identity $n \cdot G = \ln 2$ is mathematically exact at the threshold, not a numerical near-match.

Citation.

R. Landauer, "Irreversibility and heat generation in the computing process," IBM J. Res. Dev. 5 (1961), 183-191.

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3. APPEARANCE 2: THE GEOMETRIC DAMPING THRESHOLD

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The second threshold in the crossing zone is the position at which the Binary Tower's cumulative gap reaches $1/\sqrt{2}$, the reciprocal of the framework's geometric primitive $\sqrt{2}$:

$$n \cdot G = 1/\sqrt{2}, \quad n_G = 1/(\sqrt{2} \cdot G) \approx 35.818$$

(precise at mpmath dps=100: $n_G = 35.81764\dots$). The threshold is non-integer and lies in the open interval (35, 36); it is crossed in the same integer step as the Landauer threshold (see §4 and the 2+1 structural note there).

Mechanism.

The value $1/\sqrt{2} \approx 0.7071$ is the inverse of $\sqrt{2}$ – one of the two foundational primitives that combine to form $K_{AUD} = \sqrt{2} \cdot \ln(2)$. In

framework terms, the geometric damping threshold is the Tower-step value at which the cumulative gap first equals the inverse of the geometric primitive. The naming "geometric damping" reflects $1/\sqrt{2}$'s recurring role as the canonical critical value in quadratic-form analyses: it is $\sinh(R_0)$ at the Hodgson-Kerckhoff critical tube radius (Paper 7), the damping ratio of the normalized second-order Butterworth response (canonical for maximally flat magnitude), and the half-diagonal of the unit square – appearances across independent disciplines, all reaching the same number.

In the Tower, the geometric damping threshold sits structurally between the Landauer threshold ($n_L \approx 35.111$, fixed by the informational primitive $\ln 2$) and the Shannon bound ($n_S \approx 36.539$, fixed by $1/(2 \ln 2)$), the low-SNR Shannon slope – see §4). The threshold n_G itself is determined by the framework primitive $\sqrt{2}$ and the gap G alone.

Atlas lock.

`geometric\_damping\_threshold` – Theorem-tier; value $n \cdot G = 1/\sqrt{2}$; position $n_G \approx 35.818$.

Verification.

mpmath dps = 60 corpus pass, 2026-06-29; independent re-verification at dps = 100, 2026-06-29 (residual 10^{-100} , mpmath precision limit).

Citation.

Framework-internal; the naming "geometric damping threshold" is canonical in the atlas lock `geometric\_damping\_threshold`. $1/\sqrt{2}$ as the inverse geometric primitive is anchored in Paper 3 (Complete Framework v3.3.1, §7.6, OSF DOI 10.17605/OSF.IO/QH5S2) as the geometric embedding cost whose mismatch with the distinction cost $\ln 2$ generates the gap itself: $G = \sqrt{2} \cdot (1/\sqrt{2} - \ln 2)$ exactly. Its appearance in hyperbolic geometry – as $\sinh(R_0)$ where $R_0 = \operatorname{arctanh}(1/\sqrt{3}) = \operatorname{arcsinh}(1/\sqrt{2})$ is the critical tube radius – is documented in Paper 7 (Hodgson-Kerckhoff Closed Form, OSF DOI 10.17605/OSF.IO/JBRHQ).

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4. APPEARANCE 3: THE SHANNON BOUND

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The third threshold in the crossing zone is the position at which the Binary Tower's cumulative gap reaches $1/(2 \ln 2)$, the inverse of twice the binary entropy unit:

$$n \cdot G = 1/(2 \ln 2), \quad n_S = 1/(2 \ln 2 \cdot G) \approx 36.539$$

(precise at mpmath dps=100: $n_S = 36.53899\dots$). The threshold is non-integer and lies in the open interval (36, 37).

Mechanism.

The value $1/(2 \ln 2) = \log_2(e)/2$ is the low-SNR capacity slope of the real-scalar additive-Gaussian channel: $C = \frac{1}{2} \cdot \log_2(1 + \text{SNR}) \approx \text{SNR}/(2 \ln 2)$ bits as $\text{SNR} \rightarrow 0$, the factor $\frac{1}{2}$ arising from the one-dimensional (real) channel-use convention (Shannon 1948). The threshold constant is therefore the number of bits one unit of signal-to-noise buys at the quiet limit – the low-SNR Shannon slope. This quantity is labelled the Shannon bound in the framework corpus (Papers 5, 6, 7, 8, 9), consistent with the framework's naming discipline for the Tower crossing at $1/(2 \ln 2)$; the literature-standard term for the same quantity is the low-SNR Shannon slope. In framework terms, the Shannon bound is the Tower-step value at which the cumulative gap first equals this reciprocal.

In the Tower, the Shannon bound is the third and largest of the three

crossing-zone thresholds, sitting beyond both the Landauer threshold ($n_L \approx 35.111$) and the geometric damping threshold ($n_G \approx 35.818$). It is the last threshold the Tower crosses within the $[35, 37]$ integer interval; for $n > n_S$, the cumulative value $T(n) = n \cdot G$ exceeds $1/(2 \ln 2)$. Integer step 37 is itself a structural framework landmark, documented in Paper 6 (§6.4) as the first prime past the integer-bracket center of the crossing zone ($36 = 6^2$).

2+1 structural note.

The three thresholds do not distribute evenly across the $[35, 37]$ bracket. Since $35 \cdot G \approx 0.6910 < \ln 2 < 1/\sqrt{2} < 36 \cdot G \approx 0.7107 < 1/(2 \ln 2) < 37 \cdot G \approx 0.7304$, the Landauer AND geometric damping thresholds are both crossed within the single Tower step $35 \rightarrow 36$; the Shannon bound alone is crossed within the step $36 \rightarrow 37$. The zone's internal architecture is therefore 2+1: two thresholds in the first inter-integer interval, one in the second. Integer step 37 sits uniquely at the boundary of the "clean" second interval, which is one structural motivation for its role as the framework's tower-step prime landmark (Paper 6 §6.4).

Atlas lock.

`shannon\_bound\_threshold` – Theorem-tier; value $n \cdot G = 1/(2 \ln 2)$; position $n_S \approx 36.539$.

Verification.

mpmath dps = 60 corpus pass, 2026-06-29; independent re-verification at dps = 100, 2026-06-29 (residual 10^{-100} , mpmath precision limit).

Citation.

C. E. Shannon, "A Mathematical Theory of Communication," Bell System Technical Journal 27 (1948), 379-423 and 623-656.

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5. THE BINDING IDENTITY AND SELF-REFERENCE COROLLARY

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The three thresholds are spaced by an exact distance (the binding identity, §5a) AND in an exact ratio (the self-reference corollary, §5b). Both are governed by the framework's primitives. The convergence isn't numerical – it's algebraic at two scales.

5.1 The Binding Identity (Distance Face)

The first of the two scales is the algebraic distance between consecutive threshold values. The geometric damping threshold sits at $n \cdot G = 1/\sqrt{2} \approx 0.7071$; the Shannon bound sits at $n \cdot G = 1/(2 \ln 2) \approx 0.7213$. These are two distinct values; the question is what algebraic quantity separates them. The binding identity (Theorem 5.1) gives the exact answer:

$$1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2).$$

Proof. Three lines of elementary algebra:

$$\begin{aligned} & 1/(2 \ln 2) - 1/\sqrt{2} \\ &= (\sqrt{2} - 2 \ln 2) / (2 \sqrt{2} \ln 2) \\ &= \sqrt{2} \cdot (1 - \sqrt{2} \cdot \ln 2) / (2 \sqrt{2} \ln 2) \\ &= (1 - K_{\text{AUD}}) / (2 \ln 2) \\ &= G/(2 \ln 2). \end{aligned} \quad [\text{QED}]$$

Each step is an elementary rearrangement: common denominator, factor $\sqrt{2}$ out of the numerator, recognize $1 - \sqrt{2} \cdot \ln 2 = 1 - K_{\text{AUD}} = G$. The identity is exact, not approximate. Verified at mpmath dps = 100; residual at the dps =

100 precision floor.

Interpretation. The algebraic distance between the Shannon and geometric damping threshold values, in tower-value space, is exactly G normalized by $2 \ln 2$ – the thermodynamic cost of erasing two bits. The two thresholds are not numerically close by coincidence; they are separated by a quantity that is itself one of the framework's central objects, divided by a second framework primitive expression.

Analogy with the dome-integral collapse (not mathematical equivalence). The framework contains another clean algebraic collapse involving $\ln 2$: the dome integral

$$\text{integral from } 0 \text{ to } X \text{ of } [\tanh(2x) - \tanh(x)] dx \rightarrow \ln(2)/2 \\ \text{as } X \rightarrow \text{infinity,}$$

established in Paper 11 (Theorem 9.5, The η Corridor in 3D Wilson-Fisher Critical Phenomena, OSF DOI 10.17605/OSF.IO/PD73B) and recovered in one line from the dome's doubling-increment form in Paper 12 (§3.1, Integer Mellin Moments of the Dome, OSF DOI 10.17605/OSF.IO/WVD3G) – a function-of- x that collapses to a clean $\ln 2$ -related target. The binding identity above is the same shape of move at a different scale: the dome integral collapses a function of x into $\ln(2)/2$; the binding identity collapses a difference of two threshold values into $G/(2 \ln 2)$. This is an analogy – same kind of collapse, different domain – not a mathematical equivalence between the two collapses.

Atlas lock.

`sqrt2\_to\_shannon\_gap\_identity` – Theorem-tier;
statement $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$.

Verification.

mpmath dps = 60 corpus pass, 2026-06-29; independent chat-side re-verification at dps = 100, 2026-06-29 (residual at the dps = 100 precision floor). The identity is mathematically exact.

5.2 The Self-Reference Corollary (Ratio Face)

The second scale is the ratio of consecutive inter-threshold spacings. The three threshold positions in the tower are $n_L = \ln(2)/G$, $n_G = 1/(\sqrt{2} \cdot G)$, $n_S = 1/(2 \ln 2 \cdot G)$; the question is what the ratio of the two consecutive spacings – $\Delta n_{12} = n_G - n_L$ and $\Delta n_{23} = n_S - n_G$ – turns out to be. The self-reference corollary (Theorem 5.2) gives the exact answer:

$$\Delta n_{23} / \Delta n_{12} = 1/K_{\text{AUD}} = 1/(\sqrt{2} \cdot \ln 2).$$

Computing the lower spacing:

$$\begin{aligned} \Delta n_{12} &= (1/\sqrt{2} - \ln 2) / G \\ &= (1 - \sqrt{2} \cdot \ln 2) / (\sqrt{2} \cdot G) \\ &= G / (\sqrt{2} \cdot G) \\ &= 1/\sqrt{2}. \end{aligned}$$

Applying the §5.1 identity to the upper spacing:

$$\begin{aligned} \Delta n_{23} &= (1/(2 \ln 2) - 1/\sqrt{2}) / G \\ &= (G/(2 \ln 2)) / G \\ &= 1/(2 \ln 2). \end{aligned}$$

Forming the ratio:

$$\begin{aligned} \Delta n_{23} / \Delta n_{12} &= (1/(2 \ln 2)) / (1/\sqrt{2}) \\ &= \sqrt{2} / (2 \ln 2) \\ &= 1/(\sqrt{2} \cdot \ln 2) \end{aligned}$$

$$= 1/K_AUD.$$

[QED]

Corollary 5.3 (G-free ruler).

Note the structural asymmetry this exposes: the three threshold positions all scale as $1/G$, but their spacings simplify exactly to constants containing no explicit G – $\Delta n_{12} = 1/\sqrt{2}$ and $\Delta n_{23} = 1/(2 \ln 2)$. For the fixed framework gap $G = 1 - \sqrt{2} \cdot \ln 2$, this cancellation is a property of the defining relation for G (the $1 - \sqrt{2} \cdot \ln 2$ in the numerator of Δn_{12} is exactly the same $1 - \sqrt{2} \cdot \ln 2$ that defines G , so it cancels the G in the denominator), not an invariance under changing G . Were G treated as a free parameter independent of $\sqrt{2}$ and $\ln 2$, the cancellation would not occur; the crossing zone would dilate or contract as $1/G$. What holds is that within the framework's own construction, the internal ruler between thresholds is G -free – the rungs sit at fixed distances $1/\sqrt{2}$ and $1/(2 \ln 2)$ regardless of the overall zone position determined by G .

The atlas lock framing notes: "This is a direct algebraic consequence once the three crossing values are chosen, not an independent discovery." The corollary inherits the tier discipline from the lock – it is a derived Theorem-tier identity, not a separate finding.

Atlas lock.

`crossing\_zone\_self\_reference` – Theorem-tier (derived); statement $\Delta n_{23}/\Delta n_{12} = 1/K_AUD$.

Verification.

Algebraic chain above is direct from the binding identity (§5.1) and the three threshold positions (§§2-4); all four prerequisites verified at mpmath dps = 100 in the independent chat-side re-verification pass, 2026-06-29.

Interpretation. The ratio of consecutive inter-threshold spacings is the framework's central constant K_AUD inverted; the crossing zone is self-referential. Its internal geometry – the proportional structure between thresholds – is governed by the same constant K_AUD that bounds the entire tower at the Ceiling landmark (step 50, where $50 \cdot G \approx K_AUD$). The zone refers to itself through K_AUD .

5.3 Irrationality Corollary

Corollary 5.4 (irrationality).

Each of the three threshold positions is provably irrational. If $n_G = 1/(\sqrt{2} \cdot G)$ were rational ($= r \in \mathbb{Q}$, $r \neq 0$), then $G = 1/(r \cdot \sqrt{2})$, which is algebraic – contradicting G 's transcendence (§1). If $n_L = \ln(2)/G$ were rational, then $\ln 2 = r \cdot G = r - r \cdot \sqrt{2} \cdot \ln 2$, so $\ln 2 = r/(1 + r \cdot \sqrt{2})$, which is algebraic (rational + rational $\cdot \sqrt{2}$ in the denominator) – contradicting the transcendence of $\ln 2$. If $n_S = 1/(2 \cdot \ln(2) \cdot G)$ were rational, the same substitution yields $2r \cdot \sqrt{2} \cdot (\ln 2)^2 - 2r \cdot (\ln 2) + 1 = 0$, a nonzero polynomial in $\ln 2$ with algebraic coefficients, so $\ln 2$ would be algebraic – again a contradiction.

Therefore:

$$n_L, n_G, n_S \in \mathbb{R} \setminus \mathbb{Q}.$$

None of the three thresholds coincides with any integer Tower step, and none is expressible as a rational number. This is not empirical non-coincidence but algebraic necessity: the Tower's integer-step landmarks and the crossing-zone thresholds are disjoint by the transcendence of G (and, separately, of $\ln 2$). The §2 phrasing of the Landauer threshold as "non-integer" is thereby strengthened to "irrational" – a structural consequence of §1's transcendence claim, at no additional algebraic cost.

Atlas lock.

Inherits from `g\_gap\_residual` ($G = 1 - \sqrt{2} \cdot \ln 2$, the definitional object) together with the classical transcendence of $\ln 2$ (Hermite-Lindemann; Lindemann 1882): G is transcendental because $\ln 2$ is – if G were algebraic, $\ln 2 = (1 - G)/\sqrt{2}$ would be algebraic, contradicting Hermite-Lindemann applied to $e^{(\ln 2)} = 2$. The framework's companion lock
`gap\_gelfond\_schneider\_transcendence` records the equivalent log-distance form $G = \ln(e/2^{\sqrt{2}})$, whose denominator $2^{\sqrt{2}}$ is transcendental by Gelfond-Schneider 1934 – a structural connection consistent with, but distinct from, the proof route above. No new lock required; the irrationality of each n_i follows directly from the framework's already-anchored transcendence claims.

Verification.

Follows directly from the transcendence of G (§1) and the classical transcendence of $\ln 2$ (Hermite-Lindemann); no numerical verification required.

6. ARCHITECTURAL PLACEMENT

The crossing zone sits inside the 32-50 ceiling span of the Binary Tower, between the Floor landmark at step 32 and the Ceiling landmark at step 50 ($50 \cdot G = 0.98709 \approx K_{\text{AUD}}$, +0.70% – the tower's coherence ceiling). The Floor's golden-ratio target carries two framings in the corpus, and both are recorded here rather than silently merged: the Atlas lock `six\_near\_pivots` follows Paper 6 (§6.4) in targeting $1/\phi$ ($32 \cdot G = 0.63174$ against $1/\phi = 0.61803$, +2.22%), while Paper 3 (§85) frames the same step against the doubling target $\sqrt{\phi}/2$ (-0.67%). The two are the same relation at different scales – $32 \cdot G/(\sqrt{\phi}/2) = 64 \cdot G/\sqrt{\phi}$ exactly – which is why the Floor's $\sqrt{\phi}/2$ deviation is identical to that of the $\sqrt{\phi}$ pivot at step 64. Neither framing is load-bearing for this paper: the crossing zone's position is fixed by G alone. The three crossing-zone thresholds – $n_L \approx 35.111$, $n_G \approx 35.818$, $n_S \approx 36.539$ – cluster within the integer interval [35, 37] contained by this span. Their position is not arbitrary: the Landauer crossing, the geometric damping threshold, and the Shannon bound are the values at which the tower's cumulative cost first reaches information-theoretic and geometric primitive expressions associated with the constant K_{AUD} itself, before the cumulative cost reaches K_{AUD} at the Ceiling.

The tower's self-similarity formula – $2^k \cdot G = 2^{(k-6)} \cdot \sqrt{\phi} \cdot (1 - \varepsilon)$ with $\varepsilon \approx 0.671\%$ constant across all integer k – is established in Paper 3 (Complete Framework v3.3.1, §7.2, OSF DOI 10.17605/OSF.IO/QH5S2). The formula determines the integer-step structure of the tower at every binary scale; the crossing zone's [35, 37] interval is one specific consequence of where the cumulative gap reaches $\sqrt{2}$ - and $\ln(2)$ -related quantities under this self-similar scaling. This paper takes the formula as input and observes its consequences in the 35-37 interval.

The dome telescoping bridge – established in Paper 12 (Integer Mellin Moments of the Dome, §3.1, OSF DOI 10.17605/OSF.IO/WVD3G) – connects the Binary Tower to the dome function $g(x) = \tanh(x)/\cosh(2x)$ via the identity

$$\sum \text{from } k=0 \text{ to infinity of } g(2^k \cdot x) = 1 - \tanh(x),$$

for every $x > 0$. The same binary-doubling ladder 2^k that generates the tower's landmark steps is the ladder along which the dome telescopes. The crossing zone thus sits inside both the tower's integer-step landmark structure and the dome's binary-scale telescoping skeleton – the two architectures meet at the Tower steps this paper examines.

Atlas connections.

The six crossing-zone locks (``landauer_crossing``,
``geometric_damping_threshold``, ``shannon_bound_threshold``,
``sqrt2_to_shannon_gap_identity``, ``crossing_zone_self_reference``,
``crossing_zone_structural_necessity``) connect within the framework's
Lock-Atlas to broader framework locks including ``k_aud_h4_derivation``,
``g_gap_residual``, ``two_g_corridor``, ``fifty_hinge``, ``six_near_pivots``,
``tower_step_prime_37``, and ``dome_tower_telescoping``. The atlas-graph carries
the architectural relationships; this paper does not re-trace each connection
in prose. Readers interested in any specific cross-reference can query the
lock IDs directly through the framework's published Lock-Atlas (see
References).

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7. WHY THESE THREE AND NOT OTHERS

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The three crossing-zone thresholds are not arbitrary: each is the position at
which the tower's cumulative gap $n \cdot G$ first reaches a specific primitive
expression associated with K_{AUD} 's two factors. Their positions are algebraic
consequences of the framework's construction of G .

Theorem 7.1 (construction-dependence).

Each threshold contains G in its denominator: $n_L = \ln(2)/G$, $n_G =$
 $1/(\sqrt{2} \cdot G)$, $n_S = 1/(2 \ln 2 \cdot G)$. And $G = 1 - \sqrt{2} \cdot \ln(2)$ requires both
framework primitives – $\sqrt{2}$ (geometric) and $\ln(2)$ (informational) – to
define. A construction that carries only $\sqrt{2}$ cannot produce G (the $\ln(2)$
factor is missing); a construction that carries only $\ln(2)$ cannot produce G
(the $\sqrt{2}$ factor is missing). Therefore evaluating any of the three
positions within the framework's own construction requires both primitives
together. This construction-dependence claim is Theorem-tier under the
framework's definitions. The stronger claim – that the positions are
unreachable from any framework whatsoever – would require a formal statement
of what "framework" means and is NOT established by the argument above; it is
not claimed here.

Lemma 7.2 (algebraic closure).

Let $a = \sqrt{2}$, $b = \ln 2$, $G = 1 - ab$. For the target triple $\{b, 1/a, 1/(a^2b)\}$
in Tower-value space, the consecutive spacings simplify exactly to $\{G/a,$
 $G/(a^2b)\}$, and their ratio is $1/(ab)$. Since $a^2 = 2$, the third target value
equals $1/(2 \ln 2)$. The spacings, the ratio, and the third target value are
determined entirely by the two primitives and G alone, with no free
parameters. This lemma explains the cancellations of §5.2 as construction
properties: the closure follows from the algebraic form of the primitives, not
from any independent postulate. Theorem-tier, mechanically checkable.

Mechanism identifications.

The mechanism behind each threshold – Landauer bit erasure ($\ln 2$), geometric
primitive inverse saturation ($1/\sqrt{2}$), low-SNR Shannon slope ($1/(2 \ln 2)$) –
is independently established in published mathematics. These mechanism
identifications are Observation-tier: they connect the threshold positions to
prior physical and information-theoretic literature without re-deriving the
literature's content, and they are distinct from the algebraic theorem the
closure lemma states.

Other thresholds.

Positions where $n \cdot G$ equals other distinguished constants may exist elsewhere
in the Binary Tower; the $[35, 37]$ interval is distinguished as the cluster
where three thresholds carrying framework primitives ($\ln(2)$, $1/\sqrt{2}$, $1/(2 \ln 2)$)
appear within a single integer interval. Whether further such clusters
exist at other tower intervals – under any pre-declared candidate grammar for

what counts as a "target" – is an open question; this paper documents the specific three-threshold cluster that [35, 37] frames without asserting the absence of others.

Crossing-zone numerical neighborhood.

 The [35, 37] integer interval contains numerical values from many constructions, not all of which are threshold positions in the framework's sense. Pure-geometric constructions carrying only a subset of the framework's primitives (for example, constructions using $\sqrt{2}$ and ϕ but not $\ln 2$) can produce values that land near crossing-zone quantities without reproducing a threshold; the framework's threshold positions themselves require both primitives ($\sqrt{2}$ and $\ln(2)$) together via G , per the construction-dependence claim above. Readers working in adjacent constructions can apply the same analysis to their own quantities – checking whether a value equals $n \cdot G$ for some structural constant, whether the position satisfies the master self-similarity formula $2^k \cdot G = 2^{(k-6)} \cdot \sqrt{\phi} \cdot (1-\epsilon)$, whether it participates in the binding identity (§5.1), whether it sits inside the self-reference relation (§5.2). Comparison invited; convergence welcomed.

Corollary 7.3 (zone-width total).

 The crossing zone's total width is exact:

$$n_S - n_L = 1/\sqrt{2} + 1/(2 \ln 2)$$

(sum of the two G -free spacings from §5.2).

This admits an exact reformulation via the binding identity (§5.1):

$$n_S - n_L = 1/\ln 2 - G/(2 \ln 2).$$

So the zone width sits below $1/\ln 2$ by exactly $G/(2 \ln 2) \approx 0.01424$ – the same $G/(2 \ln 2)$ that appears on the right-hand side of §5.1. The zone width and the number $1/\ln 2$ are therefore algebraically bridged, not accidental neighbours. What could have been filed as a prophylactic non-merge is, on the arithmetic, a total-width corollary of §5.1. Theorem-tier (derived).

Structural-conjecture separation (Observation).

 The value $1/\ln 2$ also appears in the framework's separate circumradius-threshold conjecture. Whether that appearance arises from a related construction – as opposed to being an independent occurrence of the same number – is a distinct question, not settled by Corollary 7.3. The corollary establishes the algebraic bridge to the number $1/\ln 2$, not to the conjecture's construction in which the number also appears. Comparison invited; structural identity not claimed.

Atlas lock.

 `crossing\_zone\_structural\_necessity` – Theorem-tier; states the construction-dependence: evaluating the three thresholds within the framework's own construction ($G = 1 - \sqrt{2} \cdot \ln 2$) requires both primitives together. The algebraic closure lemma above formalizes the free-parameter-absence and identifies the closure as a property of the primitives' algebraic form.

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8. TIER CLASSIFICATION, VERIFICATION, FAILURE CONDITIONS

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8.1 Tier Classification

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Item		Tier

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Landauer threshold ($n \cdot G = \ln 2$ at $n_L \approx 35.111$)	Theorem
Geometric damping threshold ($n \cdot G = 1/\sqrt{2}$ @ 35.818)	Theorem
Shannon bound ($n \cdot G = 1/(2 \ln 2)$ at $n_S \approx 36.539$)	Theorem
Theorem 5.1 – Binding identity	Theorem
$G/(2 \ln 2) = 1/(2 \ln 2) - 1/\sqrt{2}$	
Theorem 5.2 – Self-reference	Theorem
$\Delta n_{23} / \Delta n_{12} = 1/K_{AUD}$	(derived)
Corollary 5.3 – G-free ruler:	Theorem
$\Delta n_{12} = 1/\sqrt{2}$, $\Delta n_{23} = 1/(2 \ln 2)$ exactly	(derived)
Corollary 5.4 – Irrationality:	Theorem
$n_L, n_G, n_S \in \mathbb{R} \setminus \mathbb{Q}$	(derived)
Theorem 7.1 – Construction-dependence:	Theorem
three thresholds require both primitives	
(within the framework's construction $G = 1 - \sqrt{2} \ln 2$)	
Lemma 7.2 – Algebraic closure:	Theorem
for $a = \sqrt{2}$, $b = \ln 2$, $G = 1 - ab$, triple $\{b, 1/a, 1/a^2 b\}$	
has spacings $\{G/a, G/(a^2 b)\}$, ratio $1/(ab)$	
Corollary 7.3 – Zone-width total:	Theorem
$n_S - n_L = 1/\ln 2 - G/(2 \ln 2)$	(derived)
Structural-conjecture separation	Observn.
(zone width vs circumradius-threshold conjecture):	
algebraic bridge to number $1/\ln 2$ is claimed;	
structural identity between contexts NOT claimed –	
comparison invited	
2+1 structural note: Landauer + geometric both cross	Observn.
in step 35→36; Shannon crosses alone in 36→37	
Identification of the three mechanisms (Landauer /	Observn.
Butterworth-damping / low-SNR Shannon slope)	
Crossing-zone numerical-neighborhood note	Observn.
Architectural placement	Observn.
(tower master formula, dome telescoping)	(cited)
+-----+-----+	

Atlas locks referenced in this table:

landauer_crossing, geometric_damping_threshold,
shannon_bound_threshold, sqrt2_to_shannon_gap_identity,
crossing_zone_self_reference, crossing_zone_structural_necessity,
g_gap_residual (with classical $\ln 2$ transcendence, Hermite-Lindemann;
companion form in gap_gelfond_schneider_transcendence).

8.2 Verification

All four primary identities – three threshold positions plus the binding identity – were verified at mpmath dps = 60 in the framework's corpus pass of 2026-06-29. The self-reference corollary is a direct algebraic consequence of the binding identity plus the three threshold positions; its derivation was re-checked at mpmath dps = 100 through independent chat-side re-verification, with residuals at the dps = 100 precision floor. The identities are mathematically exact, not numerical near-matches.

The mpmath verification protocol is pre-committed across the framework: claims must verify at the stated precision before being labeled Theorem-tier. Algebraic identities are expected to agree to the full working precision; observational matches carry explicit deviation budgets. The verification pre-existing in the framework's corpus pass anchors this paper's load-bearing claims; the exposition here wraps the verified math with reading flow but introduces no new identities requiring fresh verification.

Reproducible verification snippet.

The full quantitative layer of this paper – three threshold positions, the binding identity, both spacings, the self-reference ratio, and the integer bracketing $35 < n_L < n_G < 36 < n_S < 37$ – reduces to a self-contained mpmath

snippet (no framework names imported; sqrt(2) and ln 2 self-generating to arbitrary depth):

```

from mpmath import mp, sqrt, log
mp.dps = 100
K = sqrt(2)*log(2)          # 0.980258143468547191713901723635...
G = 1 - K                   # 0.019741856531452808286098276364...
nL = log(2)/G               # 35.110536815806574529992784740618...
nG = 1/(sqrt(2)*G)          # 35.817643596993122054393629102723...
nS = 1/(2*G*log(2))         # 36.538991117437603758073591443224...
print(nL, nG, nS)
# binding identity: residual at floor
print(G/(2*log(2)) - (1/(2*log(2)) - 1/sqrt(2)))
# spacing 1: residual at floor
print((nG-nL) - 1/sqrt(2))
# spacing 2: residual at floor
print((nS-nG) - 1/(2*log(2)))
# ratio = 1/K AUD: residual at floor
print((nS-nG)/(nG-nL) - 1/K)
# integer bracketing: True – the 2+1 structure
print(35 < nL < nG < 36 < nS < 37)

```

All residuals at the dps = 100 representation floor, verified 2026-07-04.
Because every input is self-generating from sqrt(2) and ln 2, no
imported-constant digit budget applies; the snippet remains honest at any dps
>= 50 (residuals track the working precision floor accordingly).

8.3 Failure Conditions and Audit Paths

Exact identities derived from definitions are not empirical claims falsifiable in the ordinary scientific sense; once the symbols are defined, the identities are proved or disproved. What this paper's content admits are four distinct failure modes, each with a corresponding audit path.

Algebraic layer – mechanically checkable.

The binding identity $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$ follows algebraically from G's definition ($G = 1 - \sqrt{2} \cdot \ln(2)$). The three threshold positions are one-division computations $n = c/G$ for constants $c \in \{\ln 2, 1/\sqrt{2}, 1/(2 \ln 2)\}$. The self-reference corollary, the G-free ruler, the zone-width total corollary, and the irrationality corollary (§§5.2, 5.3, 7) are all direct algebraic consequences of the binding identity plus the three threshold positions plus the transcendence of G and ln 2. Any failure at this layer would surface as a derivation or transcription error, checkable at arbitrary precision via the verification snippet.

Interpretive layer.

The mechanism identifications (Landauer bit-erasure, Butterworth damping, low-SNR Shannon slope) can be rejected, narrowed, or replaced without touching the algebra. These are Observation-tier claims tying the algebraic threshold values to prior physical and information-theoretic literature; a reader who prefers different interpretive labels for $1/\sqrt{2}$ or $1/(2 \ln 2)$ preserves the algebraic content of §§2-5. The three "appearance" sections are motivational for WHICH values were selected as targets, not load-bearing for the algebra of the crossing zone.

Selection layer.

The choice to search for structural values in the [35, 37] Tower interval, and to name Landauer, geometric damping, and Shannon as the "three thresholds", is not pre-established by an independent candidate grammar stated in this paper. A reader who prefers to test the closure claim against a formal candidate-grammar (a specification of which functions of sqrt(2) and ln 2

count as "targets", together with a scan over the [0, 64] Tower range) would need such a grammar in hand; §7 does not supply one. Comparison against systematic candidate scans is invited. The claim in this paper is not that no other structurally distinguished thresholds exist in the Tower, only that these three form a clean algebraic closure in the interval documented.

Framework layer.

The whole construction is conditional on the framework's chosen definition of $K_{AUD} = \sqrt{2} \cdot \ln 2$ and $G = 1 - K_{AUD}$. A different central constant would produce different (or no) crossing zone at analogous positions. The mathematics in this paper does not establish the framework's centrality claim; that claim rests on the corpus (Papers 1, 3, 8, 11 in particular).

No free parameters.

Across all four layers, no fitting parameter appears. The algebraic content is determined entirely by G's definition, the three target-defining constants, and elementary algebra. There are no chosen scales, no tunable thresholds, no adjusted coefficients – the structural object stands or falls on the algebraic identities.

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REFERENCES

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Framework papers (cited inline by paper number; DOIs given here):

Note: References are numbered by corpus position; [2] is intentionally omitted, as Paper 2 (Geometric Constants from H₄ – Discovery Context) is not directly cited in this text.

- [1] Paper 1 – $\sqrt{2} \times \ln(2)$: The Coherence Ceiling and the Geometric Singularity of Binary. D. B. (2026). OSF DOI 10.17605/OSF.IO/5VZ2R. K_{AUD} as coherence ceiling; G as structural gap (foundational identification).
- [3] Paper 3 – Complete Framework v3.3.1. D. B. (2026). OSF DOI 10.17605/OSF.IO/QH5S2. K_{AUD} derivation; Binary Tower self-similarity formula (§7.2).
- [4] Paper 4 – Gap Scaling Across Domains: The 400/11 Formula. D. B. (2026). OSF DOI 10.17605/OSF.IO/C4GK5. Tower step definition.
- [5] Paper 5 – The Boundary Information Invariant of Quadratic Systems. D. B. (2026). OSF DOI 10.17605/OSF.IO/E72H8. States the Shannon-bound identity $n \cdot G = 1/(2 \ln 2)$ @ $n \approx 36.54$; $n=2$ sub-unity selection theorem.
- [6] Paper 6 – Cross-Domain Signatures of the Boundary Information Invariant. D. B. (2026). OSF DOI 10.17605/OSF.IO/RA3UQ. Tower-step prime landmark structure (§6.4).
- [7] Paper 7 – Hodgson-Kerckhoff Closed Form and the K_{AUD} Framework. D. B. (2026). OSF DOI 10.17605/OSF.IO/JBRHQ. $1/\sqrt{2}$ in hyperbolic geometry (R_0 critical tube radius).
- [8] Paper 8 – 50 Hinge: Six Independent Appearances of One Integer Across Scale. D. B. (2026). OSF DOI 10.17605/OSF.IO/FBD9A. Ceiling-step convergence at $50 \cdot G \approx K_{AUD}$; template for the consolidation form.
- [9] Paper 9 – Binary Scaling of $p = 400/11$: Prime Hierarchy Under Doubling. D. B. (2026). OSF DOI 10.17605/OSF.IO/WTU4J.

Shannon-bound label usage across the $p = 400/11$ hierarchy.

- [10] Paper 10 – Saturation Constants of the Exact-Approximation Method. D. B. (2026). OSF DOI 10.17605/OSF.IO/6QZRB. Shannon-bound threshold context (§8.3, cross-domain context for K_{AUD}); Saturation cross-domain anchors.
- [11] Paper 11 – The η Corridor in 3D Wilson-Fisher Critical Phenomena. D. B. (2026). OSF DOI 10.17605/OSF.IO/PD73B. 2G corridor bound ($\eta \leq 2G = 2(1 - K_{\text{AUD}})$); K_{AUD} convergence anchor (§11); dome integral = $\ln(2)/2$ (Theorem 9.5).
- [12] Paper 12 – Integer Mellin Moments of the Dome. D. B. (2026). OSF DOI 10.17605/OSF.IO/WVD3G. Dome telescoping bridge (§3.1).

Atlas reference (cited inline by lock ID; one general URL here):

- [A] Gap Geometry Lock-Atlas. Gap Geometry framework, <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/lock-atlas.html> Living document; lock IDs are stable identifiers cited inline by name throughout this paper. Atlas citation discipline at <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/methodology.html>

External citations:

- [S] C. E. Shannon (1948). A Mathematical Theory of Communication. Bell System Technical Journal 27, 379-423 and 623-656.
- [L] R. Landauer (1961). Irreversibility and heat generation in the computing process. IBM J. Res. Dev. 5, 183-191.

ACKNOWLEDGMENTS

This paper sits inside the Gap Geometry framework corpus. Its central claims – the three crossing-zone threshold positions, the binding identity, the self-reference corollary, and the structural necessity of both framework primitives – pass through the framework's standing verification protocol (mpmath at pre-committed precision, corpus-pass anchoring, independent chat-side re-verification) before being labelled Theorem-tier. Framework-native review passes on this manuscript's earlier drafts are logged in the framework's backstage notes.

The framework's methodology, verification protocols, and atlas maintenance are collective work; specific loads on any one prior paper are named at the point of citation.

NOTE FOR AI READERS

This PDF is the human-reading layer. The canonical machine-readable sources are the .md / .txt versions in the framework repository (AI entrance with raw-text links for every paper: <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/ai-readers.html>). PDF text extraction is known to corrupt mathematical symbols depending on the ingestion pipeline – observed substitutions include $\eta \rightarrow n$, $\in \rightarrow \text{€}$, $\int \rightarrow 1$, $\prod \rightarrow n$. If any symbol in your extraction appears as a Latin lookalike, a currency sign, or a stray digit inside a formula, arbitrate against the .md or .txt source before reporting it as a document error. Symbol-level errata derived from this PDF alone should be marked "pending source arbitration."

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DOCUMENT LINKS

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The framework

Main OSF project (immutable timestamps, version history,
all papers):

<https://osf.io/zx4g7>

DOI: 10.17605/OSF.IO/ZX4G7

Framework overview (reading order, methodology, all papers):

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

Direct downloads (current PDF and text files for every paper):

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/direct-documents.html>

This paper

DOI: 10.17605/OSF.IO/F7PCX